EXERCISE 2

Separable Equations and Asymptotic Integrals

Original statement · ProblemsCh2.pdf

Separation of variables • Blow-up

PROBLEM STATEMENT

Solve the following equations:

(i) $\dot{x} = t^3(1-x)$ with $x(0) = 3$;
(ii) $y' = (1+y^2)\tan x$ with $y(0) = 1$;
(iii) $\dot{x} = t^2 x$ (general solution);
(iv) $\dot{x} = -x^2$ (general solution);
(v) $dy/dt = e^{-t^2} y^2$: give the solution in terms of an integral and describe the behavior as $t \to \infty$ depending on the initial condition $y(0)$. Use that $\int_0^\infty e^{-s^2}\, ds = \sqrt{\pi}/2$.

Source: ProblemsCh2.pdf, Exercise 2 (page 1).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

All five equations have the separable form $\dot{x} = g(t)\,h(x)$ (the letters $x(t), y(x)$ or $y(t)$ are used as in the statement).

- (i) $g(t) = t^3, h(x) = 1 - x$. $f(t,x) = t^3(1-x)$ and $\partial f/\partial x = -t^3$ are continuous on $\mathbb{R}^2$, so the IVP has a unique local solution (Robinson, Theorem 6.2). The only equilibrium is $x \equiv 1$; since $x(0) = 3 \neq 1$ and solutions cannot cross an equilibrium (uniqueness), $x(t) > 1$ for all t.
- (ii) $g(x) = \tan x, h(y) = 1 + y^2$. f and $\partial f/\partial y = 2y\tan x$ are continuous for $x \in (-\pi/2, \pi/2)$, which contains $x = 0$. There are no equilibria because $1 + y^2 \geq 1$.
- (iii) $g(t) = t^2, h(x) = x$: linear and homogeneous; $f, \partial f/\partial x = t^2$ continuous on $\mathbb{R}^2$. Equilibrium $x \equiv 0$.
- (iv) $h(x) = -x^2$: $f, \partial f/\partial x = -2x$ continuous on $\mathbb{R}^2$. Equilibrium $x \equiv 0$. Because f grows quadratically, finite-time blow-up is possible.
- (v) $g(t) = e^{-t^2}, h(y) = y^2$: $f, \partial f/\partial y = 2ye^{-t^2}$ continuous on $\mathbb{R}^2$. Equilibrium $y \equiv 0$. Again h is quadratic, so blow-up in finite time must be checked.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: if $\dot{x} = g(t)h(x)$ and $h(x_0) \neq 0$, then near the initial point $\frac{1}{h(x)}\dot{x} = g(t)$, and integrating both sides in t (the left side by the substitution $x = x(t)$, $dx = \dot{x}\,dt$) gives $H(x(t)) - H(x_0) = G(t) - G(t_0)$, with $H' = 1/h$ and $G' = g$. This requires $h \neq 0$ along the solution (equilibria $h(x^*) = 0$ are treated separately and are excluded from the division) and g, $1/h$ continuous (Robinson, Section 8.4).

Procedure used throughout, written explicitly each time:

- find the equilibria $h(x^*) = 0$ and the interval containing x_0 where $h \neq 0$;
- separate: $\frac{dx}{h(x)} = g(t)\,dt$;
- integrate with Barrow's rule from (t_0, x_0): $[H(s)]_{x_0}^{x} = [G(\tau)]_{t_0}^{t}$, so the constant is fixed automatically;
- solve for x, track the sign inside absolute values, and determine the maximal interval of existence.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The notes write the separable equation as $x'(t) = f(x(t))\,g(t)$, $x(t_0) = x_0$; their f is the h of this problem. If $f \neq 0$ (never), divide by f and integrate from t_0 to t; with $u = x(\tau)$, $du = x'(\tau)\,d\tau$, the limits $\tau = t_0, t$ become $u = x_0, x(t)$:

$$\int_{t_0}^{t} \frac{x'(\tau)}{f(x(\tau))}\,d\tau = \int_{x_0}^{x(t)} \frac{du}{f(u)} = \int_{t_0}^{t} g(\tau)\,d\tau.$$

'Never' is guaranteed by Picard's theorem where f and f' are continuous: a solution that reached a zero x^* of f at some t_1 would solve the same IVP as the constant $x \equiv x^*$, so it would be that constant. This is the procedure listed above; its Barrow form $[H]_{x_0}^{x} = [G]_{t_0}^{t}$ fixes the constant from the initial data, as in the logistic example of the notes.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $\dot{x} = t^3(1 - x)$, $x(0) = 3$

Equilibrium: $1 - x = 0$, i.e. $x \equiv 1$. The initial value $x(0) = 3 > 1$, so $1 - x < 0$ along the solution. Separate and integrate from $(0, 3)$ with Barrow's rule:

$$\int_{3}^{x} \frac{ds}{1 - s} = \int_{0}^{t} \tau^3\,d\tau.$$

Left side: $\int \frac{ds}{1-s} = -\ln|1 - s|$ (put $u = 1 - s$, $du = -ds$). Right side: $\left[\tau^4/4\right]_0^t = t^4/4$. Substitute the limits:

$$\Big[-\ln|1-s|\Big]_3^x = -\ln|1-x| + \ln|1-3| = -\ln|1-x| + \ln 2 = \frac{t^4}{4}.$$

Solve: $\ln|1-x| = \ln 2 - t^4/4$, hence $|1-x| = 2e^{-t^4/4}$. Because $x - 1 > 0$ for all t (no crossing of the equilibrium), $|1-x| = x - 1$ and

$$x(t) = 1 + 2e^{-t^4/4}, \qquad t \in \mathbb{R}.$$

Part (ii): $y' = (1+y^2)\tan x$**,** $y(0) = 1$

No equilibria. Separate: $\frac{dy}{1+y^2} = \tan x\, dx$. Integrate from $(0, 1)$:

$$\int_1^y \frac{ds}{1+s^2} = \int_0^x \tan\sigma\, d\sigma.$$

Left: $\arctan s$. Right: $\int \tan\sigma\, d\sigma = \int \frac{\sin\sigma}{\cos\sigma}\, d\sigma$; put $u = \cos\sigma$, $du = -\sin\sigma\, d\sigma$, giving $-\int du/u = -\ln|\cos\sigma|$. On $(-\pi/2, \pi/2)$, $\cos\sigma > 0$. Substitute the limits:

$$\arctan y - \arctan 1 = -\ln(\cos x) + \ln(\cos 0) = -\ln(\cos x).$$

With $\arctan 1 = \pi/4$ and $\ln\cos 0 = 0$:

$$\arctan y = \frac{\pi}{4} - \ln(\cos x) \quad \Longrightarrow \quad y(x) = \tan\left(\frac{\pi}{4} - \ln\cos x\right).$$

Domain: $\arctan y \in (-\pi/2, \pi/2)$ is required. Since $\cos x \le 1$, $-\ln\cos x \ge 0$, so the argument $\pi/4 - \ln\cos x \ge \pi/4 > -\pi/2$ and the constraint is $\frac{\pi}{4} - \ln\cos x < \frac{\pi}{2}$, i.e. $\ln\cos x > -\frac{\pi}{4}$, i.e. $\cos x > e^{-\pi/4}$. This defines the symmetric interval $|x| < x^*$ with

$$x^* = \arccos\left(e^{-\pi/4}\right) \approx 1.0974 \; (< \pi/2 \approx 1.5708).$$

As $x \to x^{*-}$, $\arctan y \to \pi/2$, so $y \to +\infty$: the solution blows up at the finite point x^* (and symmetrically at $-x^*$, because cos is even). The maximal interval of existence is $(-x^*, x^*)$, strictly smaller than the interval $(-\pi/2, \pi/2)$ where f is continuous. This is the situation described in Robinson, Section 6.3.

Part (iii): $\dot{x} = t^2 x$ (general solution)

Equilibrium $x \equiv 0$. For $x \neq 0$ separate and integrate (indefinite form, then fix the constant):

$$\int \frac{dx}{x} = \int t^2\, dt \quad \Longrightarrow \quad \ln|x| = \frac{t^3}{3} + c.$$

Exponentiate: $|x| = e^c e^{t^3/3}$, so $x = \pm e^c e^{t^3/3}$. Put $A = \pm e^c \in \mathbb{R} \setminus \{0\}$. The equilibrium $x \equiv 0$ corresponds to $A = 0$ and indeed satisfies the equation, so

$$x(t) = A\, e^{t^3/3}, \qquad A \in \mathbb{R},\ t \in \mathbb{R}.$$

Part (iv): $\dot{x} = -x^2$ (general solution)

Equilibrium $x \equiv 0$. For $x \neq 0$, separate: $\frac{dx}{x^2} = -dt$. Integrate:

$$\int x^{-2}\, dx = -\int dt \quad \Longrightarrow \quad -\frac{1}{x} = -t + c \quad \Longrightarrow \quad x(t) = \frac{1}{t + k}, \quad k = -c.$$

The family $x = 1/(t+k)$, $k \in \mathbb{R}$, together with the equilibrium $x \equiv 0$ (which is not obtained for any finite k), is the full set of solutions. With the initial condition $x(0) = x_0 \neq 0$ one gets $k = 1/x_0$, i.e.

$$x(t) = \frac{x_0}{1 + x_0 t}.$$

Each member with $x_0 \neq 0$ blows up at $t^* = -1/x_0$. If $x_0 > 0$ the solution exists for $t > -1/x_0$ (it decays to 0 as $t \to +\infty$). If $x_0 < 0$ it exists for $t < -1/x_0 = 1/|x_0|$ and blows up forward in time at $t^* = 1/|x_0|$.

Part (v): $dy/dt = e^{-t^2} y^2$

Equilibrium $y \equiv 0$ (initial condition $y(0) = 0$). Suppose $y_0 = y(0) \neq 0$. Separate and integrate from $(0, y_0)$:

$$\int_{y_0}^{y} \frac{ds}{s^2} = \int_0^t e^{-\tau^2}\, d\tau =: I(t).$$

Left side: $[-1/s]_{y_0}^{y} = -\frac{1}{y} + \frac{1}{y_0}$. So

$$\frac{1}{y} = \frac{1}{y_0} - I(t) \quad \Longrightarrow \quad y(t) = \frac{y_0}{1 - y_0\, I(t)}, \qquad I(t) = \int_0^t e^{-\tau^2}\, d\tau.$$

This formula also gives $y \equiv 0$ for $y_0 = 0$. The function I is odd, strictly increasing ($I' = e^{-t^2} > 0$), with $I(0) = 0$ and, by the given integral, $I(t) \to \sqrt{\pi}/2$ as $t \to +\infty$. Hence on $t \geq 0$ we have $0 \leq I(t) < \sqrt{\pi}/2$. The denominator $D(t) = 1 - y_0 I(t)$ decides the behavior.

Case $y_0 \leq 0$: then $-y_0 I(t) \geq 0$, so $D(t) \geq 1 > 0$ for all $t \geq 0$ and no blow-up occurs. As $t \to \infty$:

$$y(t) \to \frac{y_0}{1 - y_0\sqrt{\pi}/2} = \frac{2y_0}{2 - \sqrt{\pi}\, y_0} =: y_\infty \leq 0.$$

Case $0 < y_0 < 2/\sqrt{\pi}$: $y_0 I(t) < y_0\sqrt{\pi}/2 < 1$, so $D(t) > 1 - y_0\sqrt{\pi}/2 > 0$ and again $y(t) \to y_\infty = \frac{2y_0}{2 - \sqrt{\pi}\, y_0} > 0$, a finite limit larger than y_0.

Case $y_0 = 2/\sqrt{\pi}$: $D(t) = \frac{2}{\sqrt{\pi}}\left(\frac{\sqrt{\pi}}{2} - I(t)\right) = \frac{2}{\sqrt{\pi}}\int_t^\infty e^{-s^2}\, ds > 0$ for every finite t but $D(t) \to 0^+$. Thus $y(t) \to +\infty$ as $t \to \infty$, without finite-time blow-up (it grows like e^{t^2} up to a power of t, since $\int_t^\infty e^{-s^2} ds \sim e^{-t^2}/(2t)$).

Case $y_0 > 2/\sqrt{\pi}$: $D(0) = 1$ and D decreases continuously to $1 - y_0\sqrt{\pi}/2 < 0$, so there is a unique $T^* > 0$ with $I(T^*) = 1/y_0$, where $D(T^*) = 0$ and $y \to +\infty$. This is finite-time blow-up. In terms of the error function, $I(t) = \frac{\sqrt{\pi}}{2}\operatorname{erf}(t)$, so $T^* = \operatorname{erf}^{-1}\left(\frac{2}{\sqrt{\pi}\, y_0}\right)$. The solution exists only on $0 \leq t < T^*$; the limit $t \to \infty$ is not reached. Example: for $y_0 = 2$, solving $I(T^*) = 1/2$ numerically (mpmath quadrature and root finding) gives $T^* \approx 0.5510$, equal to $\operatorname{erf}^{-1}(1/\sqrt{\pi})$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitution checks:

- (i) $\dot{x} = 2e^{-t^4/4} \cdot \left(-\frac{4t^3}{4}\right) = -2t^3 e^{-t^4/4}$ and $t^3(1 - x) = t^3\left(-2e^{-t^4/4}\right)$. They agree, and $x(0) = 1 + 2 = 3$. As $t \to \pm\infty$, $x \to 1$ (the equilibrium), approached from above.
- (ii) With $\theta = \pi/4 - \ln\cos x$, $y = \tan\theta$, $y' = (1 + \tan^2\theta)\,\theta'$ and $\theta' = \tan x$. Hence $y' = (1 + y^2)\tan x$. At $x = 0$: $\theta = \pi/4$, $y = 1$. Numerically $x^* = \arccos(e^{-\pi/4}) \approx 1.0974$; at $x = x^* - 10^{-6}$ the formula gives $y \approx 5.1 \times 10^5$, confirming the blow-up.
- (iii) $\dot{x} = A\, t^2 e^{t^3/3} = t^2 x$.
- (iv) $\dot{x} = -\frac{1}{(t+k)^2} = -x^2$.
- (v) $y' = \frac{y_0 \cdot y_0 I'(t)}{(1 - y_0 I)^2} = y^2 e^{-t^2}$, since $I'(t) = e^{-t^2}$.

Time scales: in (ii) the natural variable is the spatial coordinate x, with blow-up at $x^* \approx 1.0974$ rather than as $x \to \pi/2$; in (iv) the blow-up time is $-1/x_0$; in (v) the blow-up time is $T^* = \operatorname{erf}^{-1}(2/(\sqrt{\pi}\, y_0))$ when $y_0 > 2/\sqrt{\pi}$, so for those data a statement about $t \to \infty$ is meaningless. For all other y_0 the long-time behavior is governed by the finite integral $\int_0^\infty e^{-s^2} ds = \sqrt{\pi}/2$.

SUMMARY OF (V) FOR $t \to \infty$

Initial condition	**Behavior**
$y_0 < 2/\sqrt{\pi}$	bounded, $y \to \frac{2y_0}{2-\sqrt{\pi}\,y_0}$ (equals 0 for $y_0 = 0$)
$y_0 = 2/\sqrt{\pi}$	$y \to +\infty$ as $t \to \infty$, no finite blow-up
$y_0 > 2/\sqrt{\pi}$	blow-up at finite T^* with $\operatorname{erf}(T^*) = 2/(\sqrt{\pi}\,y_0)$